Gaussian Cubic Function Report

Polynomial

$$f(q) = aq^3 + bq^2 + cq + d$$

Naive

$$g_{\text{naive}}(X) = X^3a + X^2b + Xc + d$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^3 + 3aq\sigma^2 + bq^2 + b\sigma^2 + cq + d$$

$$\begin{aligned}\mathrm{Var}(g_{\text{naive}}(X)) = \sigma^2 \big(&9a^2q^4 + 36a^2q^2\sigma^2 + 15a^2\sigma^4 + 12abq^3 + 24abq\sigma^2 + 6acq^2 \\ &+ 6ac\sigma^2 + 4b^2q^2 + 2b^2\sigma^2 + 4bcq + c^2\big)\end{aligned}$$

$$\begin{aligned}\mathrm{MSE}(g_{\text{naive}}) = \sigma^2 \big(&9a^2q^4 + 45a^2q^2\sigma^2 + 15a^2\sigma^4 + 12abq^3 + 30abq\sigma^2 + 6acq^2 \\ &+ 6ac\sigma^2 + 4b^2q^2 + 3b^2\sigma^2 + 4bcq + c^2\big)\end{aligned}$$

$$\mathrm{Bias}(g_{\text{naive}}) = \sigma^2 (3aq + b)$$

Unbiased

$$g_{\text{unb}}(X) = X^3a + X^2b - 3Xa\sigma^2 + Xc - b\sigma^2 + d$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^3 + bq^2 + cq + d$$

$$\begin{aligned}\mathrm{Var}(g_{\text{unb}}(X)) = \sigma^2 \big(&9a^2q^4 + 18a^2q^2\sigma^2 + 6a^2\sigma^4 + 12abq^3 + 12abq\sigma^2 + 6acq^2 \\ &+ 4b^2q^2 + 2b^2\sigma^2 + 4bcq + c^2\big)\end{aligned}$$

$$\begin{aligned}\mathrm{MSE}(g_{\text{unb}}) = \sigma^2 \big(&9a^2q^4 + 18a^2q^2\sigma^2 + 6a^2\sigma^4 + 12abq^3 + 12abq\sigma^2 + 6acq^2 \\ &+ 4b^2q^2 + 2b^2\sigma^2 + 4bcq + c^2\big)\end{aligned}$$

Comparison

$$\text{mean gap} = \sigma^2 (-3aq - b)$$

$$\text{variance gap} = 3a\sigma^4 \left(-6aq^2 - 3a\sigma^2 - 4bq - 2c\right)$$

$$\text{MSE gap} = \sigma^4 \left(-27a^2q^2 - 9a^2\sigma^2 - 18abq - 6ac - b^2\right)$$

$$\mathrm{Bias}(g_{\text{naive}})^2 = \sigma^4 (3aq + b)^2$$